Table 2: Overview of datasets

Verified corrected dataset variants

Dataset	Students	Skill identifiers	Records	Raw columns
ASSIST2009	4,151	136	274,590	30
ASSIST2015	19,840	100	683,801	4
Algebra 2005–2006	574	436	607,025	19

Records, students and skill identifiers are counted after cleaning; raw columns count source-file columns.
Skill identifiers are composite tags (ASSIST2009/Algebra) or sequence IDs (ASSIST2015), not uniformly atomic skills.
Algebra: labelled training subset; 436 observed composite identifiers, with 387 training IDs plus UNK.

Table 3: Training network configurations for different datasets

Validation-selected LightSAKT settings

Setting	ASSIST2009	ASSIST2015	Algebra 2005–2006
train_ratio	0.7	0.7	0.7
validation_ratio	0.1	0.1	0.1
test_ratio	0.2	0.2	0.2
learning_rate	0.0010	0.0005	0.0010
seq_len	100	100	200
optimizer	Adam	Adam	Adam
embedding_dimension	64	128	128
num_heads	2	4	4
dropout	0.1	0.1	0.2
batch_size	64	64	64
max_epochs	50	50	50
early_stopping_patience	5	5	5

LightSAKT configurations selected independently by validation AUC; target student-disjoint ratios are 70/10/20.
Adam: betas (0.9, 0.999), epsilon 1e−8, weight decay 0. Early stopping monitors validation AUC.
Split/grid seed 42; final training seeds: 42, 123, 2026, 3407, 7777. Test data do not select configurations.

Table 4: Hyperparameter tuning results for Algebra 2005–2006

Corrected causal LightSAKT · validation results

Learning Rate	Embedding	Heads	Dropout	Best validation AUC
0.000500	32	2	0.1	0.788834
0.000500	32	2	0.2	0.787610
0.000500	32	4	0.1	0.789701
0.000500	32	4	0.2	0.788728
0.000500	64	2	0.1	0.800535
0.000500	64	2	0.2	0.799574
0.000500	64	4	0.1	0.801017
0.000500	64	4	0.2	0.800187
0.000500	128	2	0.1	0.807181
0.000500	128	2	0.2	0.807049
0.000500	128	4	0.1	0.807573
0.000500	128	4	0.2	0.807942
0.001000	32	2	0.1	0.797397
0.001000	32	2	0.2	0.796752
0.001000	32	4	0.1	0.797403
0.001000	32	4	0.2	0.796880
0.001000	64	2	0.1	0.803621
0.001000	64	2	0.2	0.803011
0.001000	64	4	0.1	0.804018
0.001000	64	4	0.2	0.803866
0.001000	128	2	0.1	0.807105
0.001000	128	2	0.2	0.807842
0.001000	128	4	0.1	0.807769
0.001000	**128**	**4**	**0.2**	**0.808127**
0.005000	32	2	0.1	0.801522
0.005000	32	2	0.2	0.801106
0.005000	32	4	0.1	0.801053
0.005000	32	4	0.2	0.800365
0.005000	64	2	0.1	0.803593
0.005000	64	2	0.2	0.802481
0.005000	64	4	0.1	0.804632
0.005000	64	4	0.2	0.803808
0.005000	128	2	0.1	0.807475
0.005000	128	2	0.2	0.807133
0.005000	128	4	0.1	0.807367
0.005000	128	4	0.2	0.807673

LightSAKT; all 36 configurations. Training seed 42, split seed 42; validation AUC only.
Bold marks the highest full-precision validation AUC. Exact ties select the lexically smallest run ID.
Selected: LR 0.001, dimension 128, heads 4, dropout 0.2; epoch 26.

Table 5: Hyperparameter tuning results for ASSIST2009

Corrected causal LightSAKT · validation results

Learning Rate	Embedding	Heads	Dropout	Best validation AUC
0.000500	32	2	0.1	0.743801
0.000500	32	2	0.2	0.743286
0.000500	32	4	0.1	0.744715
0.000500	32	4	0.2	0.744041
0.000500	64	2	0.1	0.750194
0.000500	64	2	0.2	0.750272
0.000500	64	4	0.1	0.749702
0.000500	64	4	0.2	0.749725
0.000500	128	2	0.1	0.750494
0.000500	128	2	0.2	0.750732
0.000500	128	4	0.1	0.750508
0.000500	128	4	0.2	0.750191
0.001000	32	2	0.1	0.745476
0.001000	32	2	0.2	0.744907
0.001000	32	4	0.1	0.745685
0.001000	32	4	0.2	0.745398
0.001000	**64**	**2**	**0.1**	**0.750804**
0.001000	64	2	0.2	0.750408
0.001000	64	4	0.1	0.750640
0.001000	64	4	0.2	0.750068
0.001000	128	2	0.1	0.749276
0.001000	128	2	0.2	0.750502
0.001000	128	4	0.1	0.750434
0.001000	128	4	0.2	0.750732
0.005000	32	2	0.1	0.747109
0.005000	32	2	0.2	0.746947
0.005000	32	4	0.1	0.745964
0.005000	32	4	0.2	0.745198
0.005000	64	2	0.1	0.748094
0.005000	64	2	0.2	0.748185
0.005000	64	4	0.1	0.749193
0.005000	64	4	0.2	0.748850
0.005000	128	2	0.1	0.748437
0.005000	128	2	0.2	0.748855
0.005000	128	4	0.1	0.748005
0.005000	128	4	0.2	0.748094

LightSAKT; all 36 configurations. Training seed 42, split seed 42; validation AUC only.
Bold marks the highest full-precision validation AUC. Exact ties select the lexically smallest run ID.
Selected: LR 0.001, dimension 64, heads 2, dropout 0.1; epoch 24.

Table 6: Hyperparameter tuning results for ASSIST2015

Corrected causal LightSAKT · validation results

Learning Rate	Embedding	Heads	Dropout	Best validation AUC
0.000500	32	2	0.1	0.726650
0.000500	32	2	0.2	0.726148
0.000500	32	4	0.1	0.726869
0.000500	32	4	0.2	0.726163
0.000500	64	2	0.1	0.729393
0.000500	64	2	0.2	0.729126
0.000500	64	4	0.1	0.727812
0.000500	64	4	0.2	0.727373
0.000500	128	2	0.1	0.729697
0.000500	128	2	0.2	0.729193
0.000500	**128**	**4**	**0.1**	**0.730065**
0.000500	128	4	0.2	0.729873
0.001000	32	2	0.1	0.727456
0.001000	32	2	0.2	0.726489
0.001000	32	4	0.1	0.726776
0.001000	32	4	0.2	0.726751
0.001000	64	2	0.1	0.729135
0.001000	64	2	0.2	0.728457
0.001000	64	4	0.1	0.728532
0.001000	64	4	0.2	0.727659
0.001000	128	2	0.1	0.729060
0.001000	128	2	0.2	0.729033
0.001000	128	4	0.1	0.729663
0.001000	128	4	0.2	0.729658
0.005000	32	2	0.1	0.726523
0.005000	32	2	0.2	0.727004
0.005000	32	4	0.1	0.725988
0.005000	32	4	0.2	0.725784
0.005000	64	2	0.1	0.728144
0.005000	64	2	0.2	0.727962
0.005000	64	4	0.1	0.726952
0.005000	64	4	0.2	0.726402
0.005000	128	2	0.1	0.725796
0.005000	128	2	0.2	0.725489
0.005000	128	4	0.1	0.726073
0.005000	128	4	0.2	0.725331

LightSAKT; all 36 configurations. Training seed 42, split seed 42; validation AUC only.
Bold marks the highest full-precision validation AUC. Exact ties select the lexically smallest run ID.
Selected: LR 0.0005, dimension 128, heads 4, dropout 0.1; epoch 18.

Table 7: Performance comparison of knowledge-tracing models across datasets

Verified final five-seed results · GKT pending

Dataset	Metric	DKT	DKT+	GKT	DKVMN	SAKT	LightSAKT
ASSIST2009	AUC ↑	**0.757165 ± 0.000527**	0.751202 ± 0.000876	*Pending*	0.755016 ± 0.001395	0.750322 ± 0.000648	0.751056 ± 0.000766
	F1 ↑	0.818332 ± 0.000993	0.816546 ± 0.002508	*Pending*	**0.819802 ± 0.001518**	0.817401 ± 0.000943	0.816533 ± 0.001517
	Accuracy ↑	0.736135 ± 0.001022	0.733004 ± 0.001809	*Pending*	**0.736225 ± 0.001179**	0.732718 ± 0.000669	0.732229 ± 0.000801
	MSE ↓	0.189594 ± 0.000669	0.194971 ± 0.000512	*Pending*	**0.178742 ± 0.000262**	0.180373 ± 0.000252	0.180233 ± 0.000319
	BCE ↓	0.564732 ± 0.001604	0.577168 ± 0.001301	*Pending*	**0.535401 ± 0.000548**	0.539191 ± 0.000666	0.538615 ± 0.000886
ASSIST2015	AUC ↑	**0.728235 ± 0.000327**	0.720802 ± 0.001036	*Pending*	0.727605 ± 0.000989	0.725180 ± 0.000258	0.726088 ± 0.000315
	F1 ↑	0.845662 ± 0.000528	0.842588 ± 0.000766	*Pending*	**0.847155 ± 0.000456**	0.846664 ± 0.000609	0.846282 ± 0.000392
	Accuracy ↑	0.749869 ± 0.000281	0.745974 ± 0.000619	*Pending*	**0.751034 ± 0.000445**	0.749418 ± 0.000386	0.749526 ± 0.000419
	MSE ↓	0.182124 ± 0.000394	0.187262 ± 0.000415	*Pending*	**0.170776 ± 0.000228**	0.171635 ± 0.000129	0.171233 ± 0.000106
	BCE ↓	0.547349 ± 0.001032	0.559082 ± 0.001053	*Pending*	**0.515011 ± 0.000615**	0.517255 ± 0.000332	0.516098 ± 0.000340
Algebra 2005–2006	AUC ↑	0.822181 ± 0.000683	**0.823456 ± 0.000649**	*Pending*	0.817725 ± 0.001099	0.813373 ± 0.000367	0.810590 ± 0.000422
	F1 ↑	0.872422 ± 0.000891	**0.872982 ± 0.000347**	*Pending*	0.872047 ± 0.000572	0.871012 ± 0.000372	0.870356 ± 0.000350
	Accuracy ↑	0.800524 ± 0.001175	**0.801094 ± 0.000253**	*Pending*	0.798950 ± 0.000818	0.796382 ± 0.000504	0.795574 ± 0.000639
	MSE ↓	0.149683 ± 0.000862	0.152382 ± 0.000488	*Pending*	**0.141806 ± 0.000514**	0.143309 ± 0.000206	0.144069 ± 0.000364
	BCE ↓	0.466820 ± 0.002433	0.474735 ± 0.001334	*Pending*	**0.439274 ± 0.001405**	0.443098 ± 0.000598	0.445272 ± 0.001018

Final held-out test results: mean ± sample SD over seeds 42, 123, 2026, 3407, 7777, on the same locked split.
Bold marks the best completed mean per metric: higher AUC/F1/accuracy; lower MSE/BCE. Differences are descriptive.
GKT final five-seed results are pending; no partial, primary or historical GKT scores are shown. Loss_mean is labelled BCE.

Table 8: Ablation study

Corrected LightSAKT · five-seed test AUC

Architecture	ASSIST2009	ASSIST2015	Algebra 2005–2006
Default	**0.751056 ± 0.000766**	0.726088 ± 0.000315	0.810590 ± 0.000422
No Dropout	0.749340 ± 0.000716	0.725149 ± 0.000593	0.809160 ± 0.000459
No Position Encoding	0.750798 ± 0.000583	0.725749 ± 0.000183	0.811410 ± 0.000603
Single Head	0.750791 ± 0.000765	**0.726622 ± 0.000426**	**0.811755 ± 0.000448**
Without attention	0.708307 ± 0.001107	0.699283 ± 0.000443	0.772550 ± 0.000515
No layer normalization*	*Not run*	*Not run*	*Not run*
Without regularization†	0.749340 ± 0.000716	0.725149 ± 0.000593	0.809160 ± 0.000459

Held-out test AUC: mean ± sample SD over five seeds. Bold marks the highest measured mean in each dataset.

* Not run: the original image includes layer-normalization removal, but no verified experiment exists for it.

† Without regularization equals No Dropout (same evidence); it is not an independent ablation.